

Question 4 $\Rightarrow$ Online Shopping Bill Calculator

* Algorithm

- (1) Start
- (2) Input q (Quantity), p (Price), d (Discount %), t (Tax %)
- (3) If $q \leq 0$ OR $p \leq 0$ OR $d \leq 0$ OR $t \leq 0$
 Print "Invalid ~~Input~~ value entered"
 Exit Program
- (4) End If
- (5) subtotal = $q * p$
- (6) discountamount = subtotal - $((\text{subtotal} * d) / 100)$
- (7) finalbill = discountamount + $((\text{subtotal} * t) / 100)$
- (8) Print subtotal, discount amount, and final bill
- (9) End

* Pseudocode

Begin

Print "Enter quantity purchased"

Input q

Print "Enter price per item"

Input p

Print "Enter discount percentage"

Input t

If $q \leq 0$ OR $p \leq 0$ OR $d \leq 0$ OR $t \leq 0$

 Print "Error Invalid Values"

Else

 set subtotal = $q * p$

 set discountamount = subtotal - $((\text{subtotal} * d) / 100)$

 set final bill = discountamount + $((\text{subtotal} * t) / 100)$

 Print "Subtotal:", subtotal

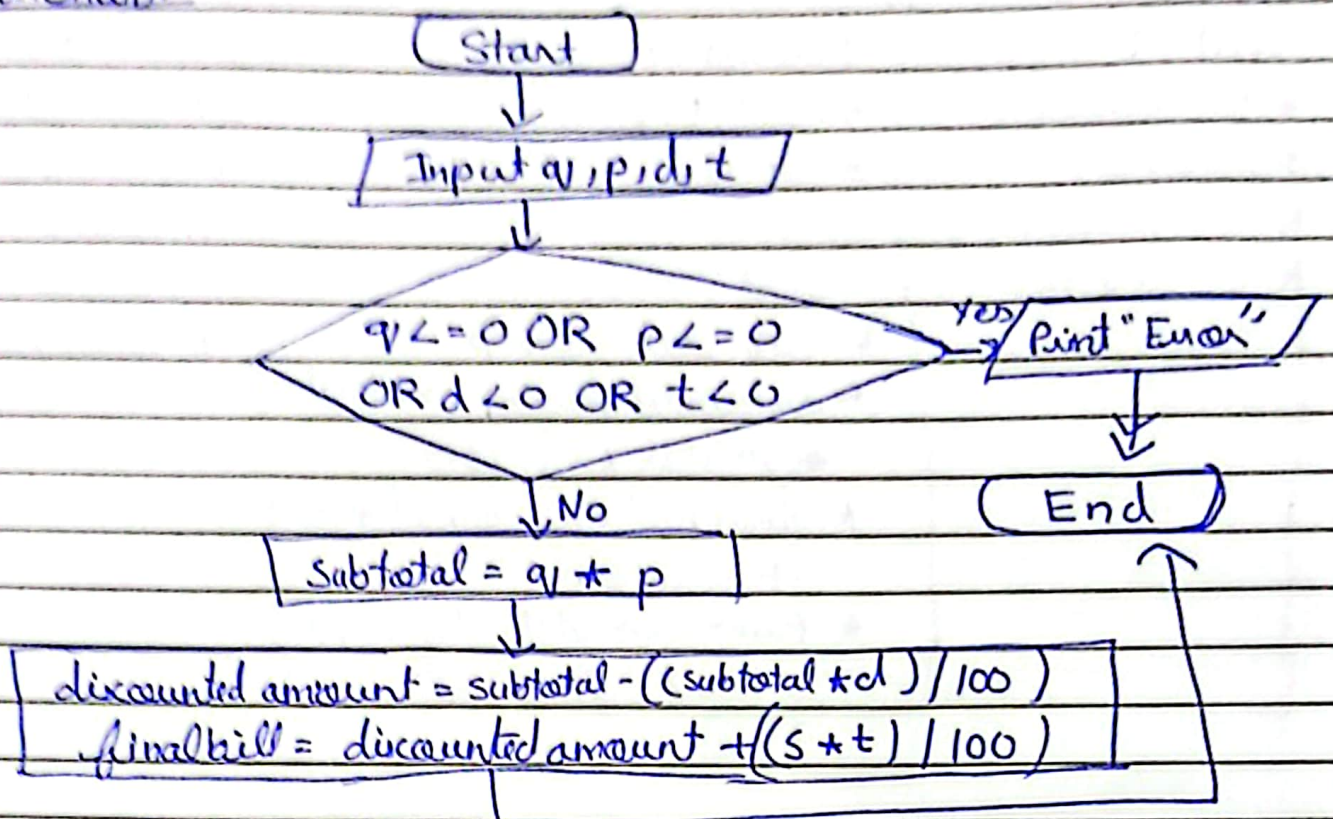
 Print "Discounted Amount:" discountamount

 Print "Final bill:" final bill

End If

End



* Flowchart* Program Analysis Chart (PAC)

Given Data	Required Processing	Output Results
* q (Quantity)	* Check if Inputs are valid	* Error message
* p (Price per item)	* Calculate subtotal,	↳ If invalid
* d (Discount %)	discounted amount and	* Bill receipt with
* t (Tax %)	final bill	subtotal
* s = q * p	* Print error if invalid,	* Discount, tax
* s = (s * d) / 100	otherwise display receipt	and final Bill

* IPO chart

INPUT	Processing	Output
* q (quantity)	* Input q, p, d, t	* subtotal
* p (price)	* If $q < 0$ OR $p < 0$	* discounted
* d (discount %)	OR $d < 0$ OR $t < 0$	amount
* t (tax %)	Print "error"	* tax amount
	* $subtotal = q * p$	* final bill
	* $discounted\ amount = subtotal -$	
	$(subtotal * d) / 100$	
	* $final\ bill = discounted\ amount$	
	$+ (subtotal * t) / 100$	
	* Print subtotal, discounted	
	amount, final bill	

